Summary in Graph

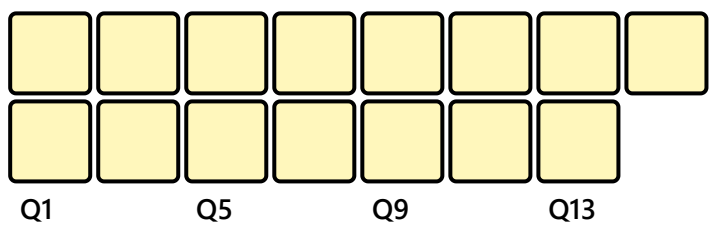
Exam Summary (GO Classes CS Test Series 2026 | C-Programming | Topic Wise Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (237) out of 2370 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Programming in C

Consider the k bit binary pattern. $\mathrm{T}_{\max}$ and $\mathrm{T}_{\min}$ are maximum and minimum signed numbers we can represent using k bits. $\mathrm{U}_{\max}$ and $\mathrm{U}_{\min}$ are maximum and minimum unsigned numbers we can represent using k bits. Numbers are represented using 2's complement system.

Which of the following(s) is/are true for $k = 16$?

Here $|.|$ represents absolute value of a number i.e. $|r| = -r$ if $r < 0$ otherwise $|r| = r$.

A. $|\mathrm{T}_{\min}| = \mathrm{T}_{\max} + 1$
B. $\mathrm{U}_{\max} = 2 * \mathrm{T}_{\max} + 1$
C. $\mathrm{U}_{\max} = |\mathrm{T}_{\min}| + \mathrm{T}_{\max} + 1$
D. $\mathrm{U}_{\min} = |\mathrm{T}_{\min}|$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Programming in C

Consider Figure A and Figure B, which represent 4 bit signed and unsigned numbers, respectively, in the $2's$ complement system.

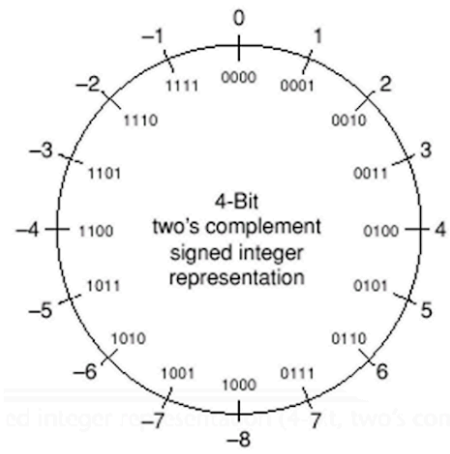


Figure A

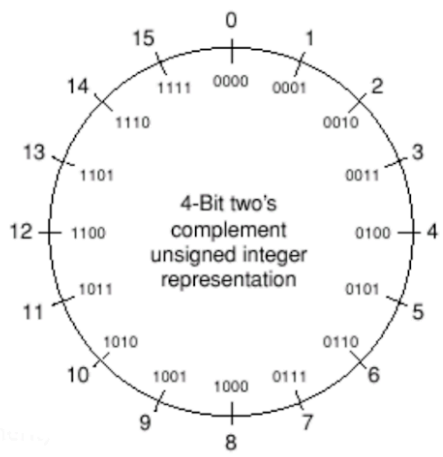


Figure B

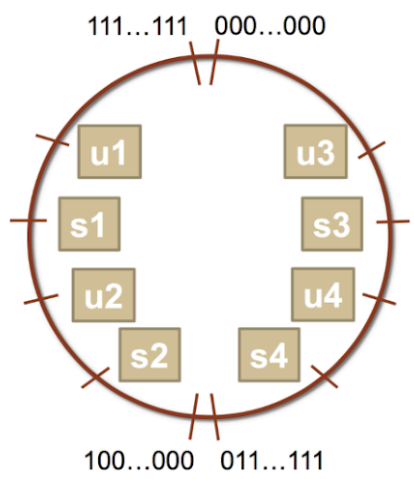


Figure C

Assume that a few variables are defined below and initialized in such a way that places them in the spot shown in Figure C.

```
int s1, s2, s3, s4;
unsigned int u1, u2, u3, u4;
```

Which of the following(s) is/are FALSE?

A. $s2 > s4$
B. $s1 > s2$
C. $s1 > u3$
D. $s3 > u3$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Assume that x is a two's complement binary integer whose absolute value is not large (so no overflow problems). At right are four C expressions that use the $<<$ (shift left) and $\sim$ (bitwise complement: flip each 0 to 1 and each 1 to 0 in the binary representation) operators. For example, $7 << 1$ is 14 and ~ 0 is -1.

Match each expression to one of the mathematical functions at left.

$a.$	$4x$	I.	$(\sim x) + 1$
$b.$	$-x$	II.	$(x << 1) + x$
$c.$	$3x$	III.	$\sim x$
$d.$	$-x - 1$	IV.	$x << 2$

A.
a - IV
b - I
c - II
d - III

B.
a - IV
b - III
c - II
d - I

C.
a - II
b - I
c - IV
d - III

D. None of these

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be the output of the following program?

```c
#include<stdio.h>
main()
{
    char grade='A';
    int result=0;
    switch(grade)
    {
        case 'A':
            result+=4;
        case 'B':
            result+=3;
        case 'C':
            result+=2;
        default:
            result+=1;
    }
    printf("%d", result);
}
```

A. 10
B. 4
C. 9
D. 0

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What is the logical condition under which the following while loop will terminate?

```c
int Beta = 5;
while (Beta > 0 && Beta < 10)
{
    printf("%d", Beta);
    scanf("%d", &Beta);
}
```

A. Beta < 0 && Beta $>= 10$
B. Beta < 0 || Beta > 10
C. Beta $<= 0$ || Beta $>= 10$
D. Beta < 0 || Beta $>= 10$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be output on the execution of the following code segment?

```
#include<stdio.h>
main()
{
    unsigned num1 = 1;
    signed num2 = -1;
    if (num1 < num2)
        printf("less");
    else if (num1 > num2)
        printf("greater");
    else if (num1 == num2)
        printf("equal");
}
```

A. greater
B. less
C. equal
D. Error

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Imagine that our system uses a 5-bit integer representation and does addition and subtraction using the rules for 5-bit, two's complement arithmetic.

Which of the following is/are TRUE?

When converting to decimal, you should treat the value as either signed or unsigned according to the rules of C, where $\mathrm{T_{min}}$ and $\mathrm{T_{max}}$ are signed, as are plain constants, but a constant ending in U is unsigned.

A. -$\mathrm{T_{max}} - 1\mathrm{U}$ is 16 in decimal
B. -$\mathrm{T_{max}} - \mathrm{T_{min}}$ is 00001 in binary
C. $\mathrm{T_{max}} + \mathrm{T_{max}}$ is -2 in decimal
D. $\mathrm{T_{min}} - 1\mathrm{U}$ is 15 in decimal

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output of the given program?

```c
#include<stdio.h>
void main()
{
    int sum =1;
    for ( unsigned int i=3; i>=0; --i)
    {
        if(i==0) sum = sum+1;
        else sum = sum *i;
    }
    printf("%d", sum);
}
```

A. 1
B. 7
C. 0
D. None of these

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 02sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming

Which of the following condition(s) is/are evaluated to true for a given declaration of integer i.

```c
int i = -7;
```

A. $-10 < i < -1$
B. $i \ \& \ 0 \mid 0 \ \& \ 1$
C. $0 \ \& \ 0 \ \& \ 0 \mid i$
D. $!(1 > 0 > i \ \& \ !i)$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

The statement

```c
while ( --counter >= 1 )
    counter % 2 ? printf("A") : printf("B");
```

can NOT be rewritten as

A.

```c
while ( --counter >= 1 )
    if ( counter % 2 )
        printf("A");
    else
        printf("B");
```

B.

```c
while ( counter >= 1 )
    if (counter % 2)
        printf("A");
    else
        printf("B");
    --counter;
```

C.

```
    while ( counter > 1 )
    {
        --counter;
        if ( counter % 2 )
5.          printf("A");
        else
            printf("B");
    }
```

D.

```
    do
    {
        --counter;
        printf( counter % 2 ? "A" : "B" );
5. } while ( counter >= 2 );
```

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider the following program fragment. Which of the following if condition(s) prints GO Classes?

```
    int b = 1, c = 1, d = 0;
    if (0 && 0 == 0) //first if
        printf("GO Classes");
    if (b || --b == 0) //Second if
5.      printf("GO Classes");
    if (c || c-- == 0) //Third if
        printf("GO Classes");
    if (d || ++d == 0) //Fourth if
        printf("GO Classes");
```

A. First if condition
B. Second if condition
C. Third if condition
D. Fourth if condition

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Let A and B be two unknown 8-bit $2's$ complement numbers. We know the results of $\mathrm{A} \wedge \mathrm{B}$ and $\mathrm{A} \,\&\, \mathrm{B}$ as shown below:

$\mathrm{A} \wedge \mathrm{B}$	00110100
$\mathrm{A} \,\&\, \mathrm{B}$	11001001

What is the sum $\mathrm{A} + \mathrm{B}$ expressed in the 8-bit two's complement notation?

A. 11000110
B. 11000010
C. 01000010
D. 11010110

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Which of the following if statement(s) below is/are equivalent to the given switch statement (that is, produces the same output under the same conditions)? Assume the **answer** is a previously declared int.

```
switch (answer)
{
    case 0: printf("0 entered"); break;
    case 1: printf("1 entered"); break;
    case 3: printf("3 entered"); break;
    case 5: printf("5 entered"); break;
    default: printf("Other value entered");
}
```

A.

```
if (answer == 0 || 1 || 3 || 5)
    printf("%d entered", answer);
else
    printf("Other value entered");
```

B.

```
if ((answer >= 1 && answer <= 5 &&
        answer % 2 == 1) || (answer == 0))
    printf("%d entered", answer);
else
    printf("Other value entered");
```

C.

```
if (answer == 0)
    printf("0 entered");
if (answer == 1)
    printf("1 entered");
if (answer == 3)
    printf("3 entered");
if (answer == 5)
    printf("5 entered");
if (answer != 0 && answer != 1
        && answer != 3 && answer != 5)
    printf("Other value entered");
```

D.

```
if (answer == 0)
    printf("0 entered");
else if (answer == 1)
    printf("1 entered");
else if (answer == 3)
    printf("3 entered");
else if (answer == 5)
    printf("5 entered");
else if (answer != 0 & 1 & 3 & 5)
    printf("Other value entered");
```

Your Answer: | **Correct Answer: B;C;D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in C

Consider two given C programs. Which of the following is/are true? It is also known that in C programming, && has higher precedence than ||.

```
Program 1
#include<stdio.h>
void main()
{
    int i = 0;
    if (++i && (i==0))
        printf("GO Classes\n");
    else
        printf("GATEOverflow\n");
}
```

```
Program 2
#include<stdio.h>
void main()
{
    int x, y, z;
    x = 1;
    y = z = 0;
    ++x || ++y && ++z;
    if (y || z)
        printf("GO Classes\n");
    else
        printf("GATEOverflow\n");
}
```

A. Program 1 prints GO Classes
B. Program 1 prints GATEOverflow
C. Program 2 prints GO Classes
D. Program 2 prints GATEOverflow

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output of the following C program?

```
#include<stdio.h>
int main(){
int x=3;
int y;
switch(x++){
    x++;
    case 3:
        printf("Three %d",x);
        break;
    case 4:
        printf("Four %d",x);
        break;
    case 5:
        printf("Five %d",x);
        break;
    default: printf("Default %d",x);
    }
}
```

A. Three 3
B. Four 4
C. Three 4
D. Four 5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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